



Intelligent Transportation Systems (Estonia)

Merika Mote



Estonia: A Digital-First Nation Driving

Smart Mobility

- ▶ Estonia stands at the forefront of digital innovation, seamlessly integrating its advanced digital society with cutting-edge transportation solutions.
 - ▶ First country to legalize autonomous vehicle testing on public roads (2016), demonstrating a proactive approach to future mobility.
 - ▶ Seamless integration of digital society with transport through e-ID, digital twins, and AI-powered traffic management.
 - ▶ Our vision: To achieve safer, smoother, and sustainable transport for all through continuous innovation and digitalization.
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Smart Infrastructure Revolutionizing Traffic Safety and Flow

- **Smart Crosswalks**
- Bercman Technologies' innovative pedestrian crosswalks communicate directly with self-driving vehicles, enhancing safety for vulnerable road users.
- **e-Pavement Sensors**
- Embedded e-Pavement sensors provide real-time data on road conditions, enabling proactive maintenance and dynamic traffic management.
- **Digital Twin Technology**
- A nationwide digital twin provides cost-effective 3D data collection and AI analysis for unparalleled traffic optimization.



Autonomous Vehicles: From Last-Mile Delivery to Public Transport

- **Starship Technologies**
- Autonomous delivery robots are active in Tallinn and globally, revolutionizing parcel delivery and logistics.
- **AuveTech Shuttles**
- Hydrogen-powered self-driving shuttles seamlessly integrate into public transport networks, particularly in Tartu.
- **Clevon & Milrem**
- Clevon's autonomous delivery trucks and Milrem Robotics' unmanned ground vehicles serve commercial and defense applications.



AI and Data-Driven Traffic Management

- ▶ Estonia leverages advanced AI and robust data analytics to optimize traffic flow and enhance road safety across its networks.
- ▶ **Tark Tee System:** Provides real-time road conditions, weather forecasts, and traffic restrictions, seamlessly integrated with navigation apps like Waze.
- ▶ **AI Platforms:** Analyze metropolitan traffic patterns to reduce congestion, predict bottlenecks, and improve overall safety through intelligent routing.
- ▶ **Mobile Mapping:** On-demand systems process vast data streams from multiple sensors, offering dynamic and actionable traffic insights.



Estonia's Global Leadership and Collaboration in ITS

- ▶ Estonia actively engages with the international community, fostering collaboration and showcasing its pioneering work in ITS.
- ▶ **Global Showcases**
- ▶ Active participation and presentations at prestigious events like the ITS World Congress 2024 in Dubai and Hamburg.
- ▶ **Research Partnerships**
- ▶ Collaborations with leading research centers, including TalTech and Florida Polytechnic University, to enhance autonomous vehicle security.
- ▶ **ITS Estonia Platform**
- ▶ This platform fosters critical cooperation between public and private sectors, driving innovation and facilitating the export of smart mobility solutions.



Thank You